Problem Set 3

Student handout

Clock arithmetic, Fermat's little theorem, and repeating continued fractions

Week 3 | 100 points + 10 optional challenge points

Name: _____ **Due:** _____

Instructions. Work exactly: no decimal approximations. When you need the floor of a surd, justify it with integer bounds such as $2 < \sqrt{7} < 3$. Justify every proof or counterexample. You may use Bézout's identity, Euclid's lemma in the form "if $n \mid cm$ and $\gcd(n, c) = 1$, then $n \mid m$," and the fact from Problem Set 2 that coprime divisors of c have a product dividing c.

Facts from lecture. *Fermat's little theorem:* if p is prime, then $a^p \equiv a \pmod{p}$ for every integer a, and $a^{p-1} \equiv 1 \pmod{p}$ when $p \nmid a$. *Integer states:* if $x_n = (P_n + \sqrt{D})/Q_n$ with $Q_n \mid D - P_n^2$ and $a_n = \lfloor x_n \rfloor$, then

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = \frac{D - P_{n+1}^2}{Q_n}, \qquad x_{n+1} = \frac{P_{n+1} + \sqrt{D}}{Q_{n+1}}.$$

Galois: a quadratic surd x has a purely periodic continued fraction exactly when it is *reduced*: $x > 1$ and $-1 < x' < 0$, where x' is the conjugate. *Reversal:* if $x = [\overline{b_0, \ldots, b_{\ell-1}}]$, then $-1/x' = [\overline{b_{\ell-1}, \ldots, b_0}]$.

I. Congruences and inverses

1. Warming up the clock

[10 points]

(a) Find the least nonnegative residue of -100 modulo 13. (2)

(b) Find the last two digits of $99 \cdot 101 \cdot 103$ without multiplying the numbers out. (2)

(c) Prove that every square is congruent to 0 or 1 modulo 4. Deduce that no integer congruent to 3 modulo 4 is a sum of two squares. Is 2027 a sum of two squares? (6)

2. Inverses by Euclid

[14 points]

(a) Use the extended Euclidean algorithm to find the inverse of 23 modulo 89. Check your answer by multiplication. (5)

(b) Solve $23x \equiv 5 \pmod{89}$. (2)

(c) Let $n \geq 2$. Prove that an inverse modulo n, when it exists, is unique modulo n: if $ab \equiv 1$ and $ac \equiv 1 \pmod{n}$, then $b \equiv c \pmod{n}$. (3)

(d) List the units modulo 15, and give the inverse of each. Which units are their own inverses? (4)

3. Cancellation, square roots of one, and Wilson's theorem

[14 points]

(a) Prove the cancellation law: if $ac \equiv bc \pmod{n}$ and $\gcd(c, n) = 1$, then $a \equiv b \pmod{n}$. Give an example showing that the conclusion can fail when $\gcd(c, n) > 1$. (5)

(b) Let p be prime. Prove that $a^2 \equiv 1 \pmod{p}$ forces $a \equiv \pm 1 \pmod{p}$. Where does your proof break for the modulus 8? (4)

(c) Pair each of the residues $2, 3, \ldots, 9$ with its inverse modulo 11, and use the pairing to compute $10!$ modulo 11 with almost no arithmetic. Then prove *Wilson's theorem*: $(p-1)! \equiv -1 \pmod{p}$ for every prime p. (5)

II. Fermat's little theorem

4. Large powers made small [12 points]

(a) Compute $3^{1000} \pmod{13}$. (3)

(b) Compute $5^{2026} \pmod{23}$. (3)

(c) Let p be prime and $p \nmid a$. Explain why a^{p-2} is an inverse of a modulo p. Use this to find the inverse of 7 modulo 19 by repeated squaring, and check your answer by multiplication. (6)

5. Fermat as a proof tool, and a warning [14 points]

(a) Prove that $30 \mid n^5 - n$ for every integer n. (5)

(b) Let p be a prime other than 2 and 5. Prove that p divides infinitely many of the numbers 9, 99, 999, 9999, … (4)

(c) Verify that $2^{10} = 3 \cdot 341 + 1$, and deduce that $2^{340} \equiv 1 \pmod{341}$. Yet $341 = 11 \cdot 31$ is not prime. Does this contradict Fermat's theorem? Finally, use Fermat modulo 31 to show that $3^{340} \not\equiv 1 \pmod{341}$, and explain how this congruence alone proves that 341 is composite. (5)

III. Continued fractions of quadratic surds

6. Running the integer-state algorithm [14 points]

(a) Make a table of n, P_n, Q_n, x_n, a_n for $x_0 = \sqrt{7}$, continuing until a state (P_n, Q_n) repeats. State the continued fraction of $\sqrt{7}$. (6)

(b) Let $x = (1+\sqrt{2})/3$. Explain why $(P_0, Q_0) = (1, 3)$ with $D = 2$ is not a legal starting state. Show that $x = (3+\sqrt{18})/9$ is legal, and run the algorithm to find the continued fraction of x. (6)

(c) Which of the complete quotients $x_0, x_1, x_2, \ldots$ in (b) are reduced? Compare with where the period begins. (2)

7. From a period back to a surd [12 points]

(a) Let $x = [\overline{3, 1}]$. Find a quadratic equation with integer coefficients satisfied by x, and write x in the form $(P + \sqrt{D})/Q$. Confirm directly that x is reduced. (5)

(b) Use the reversal corollary to find $t = [\overline{1, 3}]$ in closed form, and then find $y = [2; \overline{1, 3}]$. (4)

(c) Your answers should satisfy $y = x - 1$. Explain this by looking only at the digits. Is y purely periodic? Reconcile your answer with Galois's theorem. (3)

8. Galois's test and two infinite families [10 points]

(a) Without computing any continued fractions, decide which of

$$2 + \sqrt{7}, \qquad \frac{3+\sqrt{5}}{2}, \qquad \frac{2+\sqrt{7}}{3}, \qquad \frac{1+\sqrt{7}}{4}$$

have purely periodic continued fractions. Then read off the expansions of the purely periodic ones from your table in Problem 6(a). (4)

(b) Let $n \geq 1$. Prove that $\sqrt{n^2+1} = [n; \overline{2n}]$. (3)

(c) Let $n \geq 1$. Find and prove a formula for the continued fraction of $\sqrt{n^2+2}$. (3)

Optional challenge: the period of $\sqrt{D}$ is a palindrome [10 extra points]

Let D be a positive nonsquare integer and $a_0 = \lfloor \sqrt{D} \rfloor$. Prove that

$$\sqrt{D} = [a_0; \overline{a_1, a_2, \ldots, a_{\ell-1}, 2a_0}] \qquad \text{with} \qquad a_i = a_{\ell-i} \quad (1 \leq i \leq \ell - 1).$$

Compare with $\sqrt{7}$ in Problem 6. *Hint:* apply Galois's theorem and the reversal corollary to $x = a_0 + \sqrt{D}$. You may use the fact that an irrational number has only one simple continued fraction expansion.